

Lubna Zahan Lamia

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Research Assistant, Cognitive Agents and Interaction Lab (CAIL) — University of Dhaka, Bangladesh

RESEARCH PROFILE

My research investigates how **large language models** construct, reshape, and mediate meaning at the intersection of **natural language processing**, **computational linguistics**, and **ethical AI**. I focus on uncovering how bias, perspective, and interpretive reliability manifest in generative text, integrating linguistic theory with computational analysis to examine how modern language technologies transform patterns of meaning and representation across linguistic structure and model behavior.

RESEARCH INTERESTS

Natural Language Processing • Computational Linguistics • Deep Learning

Bias and Perspective in NLP • Ethical and Trustworthy AI • Responsible Language Technologies

EDUCATION

University of Dhaka

Bachelor of Science in Computer Science & Engineering

- Cumulative GPA: 3.53 / 4.00 (Final semester GPA: 3.89 / 4.00, ranked 3rd of 70)

Dhaka, Bangladesh

Jan 2020 – July 2025

Holy Cross College

Higher Secondary Certificate (Science)

- Result: GPA 5.00 / 5.00; awarded Government Scholarship (H.S.C.).

Dhaka, Bangladesh

June 2017 – June 2019

Savar Cantonment Public School and College

Secondary School Certificate (Science)

- GPA: 5.00 / 5.00; awarded Government Scholarship (S.S.C.).

Dhaka, Bangladesh

Completed 2017

RESEARCH EXPERIENCE

Research Assistant

Cognitive Agents and Interaction Lab (CAIL), University of Dhaka

- Investigating human–AI collaboration dynamics in generative systems, with a focus on how humans and large language models co-construct meaning and interpretive outcomes.
- Conducting research on bias, framing, and interpretability in large language models.
- Developed frameworks for analyzing narrative distortion and sentiment-informed modeling across generative systems.
- Contributed to peer-reviewed research works bridging computational linguistics and deep learning.

Jan 2024 – Present

Dhaka, Bangladesh

PUBLICATIONS

Published / Accepted

- Lamia, L. Z.**, Hossain, M. F. B., & Khan, M. M. (2025). *Who Holds the Pen? Caricature and Perspective in LLM Retellings of History*. *Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025*, Suzhou, China — **Oral Presentation**, November 5–9, 2025; **Outstanding Paper (SAC Highlight) Award Nominee**.

Under Review

- Hossain, M. F. B., **Lamia, L. Z.**, & Khan, M. M. (2025). *FinBERT-BiLSTM: A Deep Learning Model for Predicting Volatile Cryptocurrency Market Prices Using Market Sentiment Dynamics*. Under review at *Applied Intelligence Journal (Springer)* — Equal Contribution.

ONGOING & FUTURE RESEARCH PROJECTS

Emerging Work on Context-Aware Evaluation in Large Language Models

- Extending prior narrative-level analyses toward more grounded, multimodal, and culturally adaptive evaluation frameworks.

Extension of “Who Holds the Pen? Caricature and Perspective in LLM Retellings of History”

- Expanding the EMNLP 2025 study to cross-domain analyses of interpretive bias and perspective consistency.

Extension of FinBERT-BiLSTM Hybrid Model for Cryptocurrency Forecasting

- Developing a real-time platform for sentiment-driven market forecast visualization and evaluation.

TECHNICAL SKILLS

Python, C++, Java | PyTorch, TensorFlow, Scikit-learn | Hugging Face (FinBERT, Sentence-BERT, RoBERTa-MNLI, LLaMA) | OpenAI GPT series, Claude, Gemini | Bi-LSTM, TCN, TFT, Informer | Git, Google Cloud

AWARDS & RECOGNITIONS

- **EMNLP 2025 Diversity & Inclusion Grant** recipient (USD 1000).
- **Full registration waiver** (USD 600) for EMNLP 2025 participation, including ACL membership.
- **Outstanding Paper (SAC Highlight) Award Nominee** — selected as one of only **75 papers** across all award categories out of **8,174** EMNLP 2025 submissions.
- **World Youth Leadership Conference (WYLC) 2025**, Dubai — awarded partial scholarship from Global Entrepreneurship Bootcamp INC.
- **Government Scholarships** for academic merit in H.S.C., S.S.C., and J.S.C. examinations.

CO-CURRICULAR & LEADERSHIP ACTIVITIES

- Served as a **Student Volunteer** at the *Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025*, Suzhou, China — In-person.
- **Departmental Coordination** — **Dept. of CSE, University of Dhaka** Led organization of *Graduation Week 2024*, coordinating multi-day events, scheduling, registration, and volunteer activities.
- **Program Host** — **University of Dhaka** Served as host for events including the *Graduation Ceremony*, *Freshers' Reception*, and *CAIL LLM Workshop*, supporting program flow and on-site coordination.

ACADEMIC REFERENCES

Dr. Md. Mosaddek Khan

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Department of Computer Science and Engineering
University of Dhaka, Bangladesh
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Mabsur Fatin Bin Hossain

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